

Finding the Mistake in Sums of Radicals

Answer Key

Grade 8 / Pre-Algebra

Quick Answers

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| 1. (a) the sum of the two roots = 7, (b) the root of the sum = 5, (c) the comparison = > | 5. (a) the sum of the two roots = 3.15, (b) the root of the sum = 2.24, (c) the comparison = > |
| 2. (a) the corrected value = 14, — | 6. $5\sqrt{2}$ |
| 3. $8\sqrt{2}$ | 7. (a) the corrected value = $6\sqrt{3}$, — |
| 4. $5\sqrt{5}$ | 8. (a) the simplified value = $7\sqrt{2}$, — |
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What is verified

5 of 8 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 2, 7, 8. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 8 / Pre-Algebra

8.NS.A – *problems 1, 2, 3, 4, 5, 6, 7, 8*

Difficulty 1–5 of 5 across 15 tagged checks.

Common wrong answers

2. *If they got 10: added the two numbers under the radicals and took a single root*

1. Test the shortcut: (a) $\sqrt{9} + \sqrt{16}$, (b) $\sqrt{9 + 16}$, (c) compare them.

Solution (a): Each radicand is a perfect square, so evaluate the roots first and then add: $3 + 4$. 7

Solution (b): Here the addition happens *under* the radical first: $9 + 16 = 25$, and the root of that is a whole number. 5

Solution (c): Seven is larger than five, so the two sides are not equal — the radical did not distribute across the plus sign. >

2. Priya writes $\sqrt{36} + \sqrt{64} = \sqrt{36 + 64} = 10$. (a) Correct value? (b) What did she do wrong?

Solution (a): Evaluate each root separately, then add: $6 + 8$. 14

Solution (b) — instructor-judged. Priya added the two numbers underneath the radical signs, as though the radical distributed across the addition. It does not: the root of

a sum is not the sum of the roots, and part (a) is the proof — the correct total is four larger than the value she wrote. A full-credit answer states that rule in some form and points at the numerical difference.

3. Combine into a single radical term: $3\sqrt{2} + 5\sqrt{2}$.

Solution: Both terms carry the same radical part, so they are like terms. Add the counts in front and carry the radical along unchanged. $8\sqrt{2}$

4. Combine into a single radical term: $7\sqrt{5} - 2\sqrt{5}$.

Solution: Like terms again, so subtract the counts in front and leave the radical part exactly as it was. $5\sqrt{5}$

5. Test the shortcut with decimals: (a) $\sqrt{2} + \sqrt{3}$, (b) $\sqrt{5}$, (c) compare.

Solution (a): Neither root is exact, so round each to two places and add: $1.41 + 1.73$, which to the nearest hundredth from the unrounded values gives 3.15

Solution (b): Squeezing between $2.23^2 = 4.9729$ and $2.24^2 = 5.0176$ traps the radicand. 2.24

Solution (c): The sum of the two roots is about nine tenths larger, so once again the shortcut fails — this time with radicands that are not perfect squares at all. $>$

6. Simplify $\sqrt{8} + \sqrt{18}$ into a single radical term.

Solution: Pull out the largest perfect-square factor of each radicand.

$$\begin{aligned}\sqrt{8} &= \sqrt{4} \cdot \sqrt{2} = 2\sqrt{2} \\ \sqrt{18} &= \sqrt{9} \cdot \sqrt{2} = 3\sqrt{2}\end{aligned}$$

Both terms now carry the same radical part, so the counts in front add. $5\sqrt{2}$

7. Marco writes $4\sqrt{3} + 2\sqrt{3} = 6\sqrt{6}$. (a) Correct value? (b) What did he do wrong?

Solution (a): The radical parts already match, so add only the counts in front: four of them plus two of them. $6\sqrt{3}$

Solution (b) — instructor-judged. Marco added the numbers under the radicals as well as the counts in front. The radical part is the shared *name* of the two terms; it is what makes them like terms, and it must come through unchanged. Only the counts in front may be added. A full-credit answer says explicitly that the radical part must not change.

8. **Challenge.** Simplify $\sqrt{50} + \sqrt{32} - \sqrt{8}$, and justify why the terms combine.

Solution (a): Every radicand hides a perfect-square factor.

$$\begin{aligned}\sqrt{50} &= \sqrt{25} \cdot \sqrt{2} = 5\sqrt{2} \\ \sqrt{32} &= \sqrt{16} \cdot \sqrt{2} = 4\sqrt{2} \\ \sqrt{8} &= \sqrt{4} \cdot \sqrt{2} = 2\sqrt{2}\end{aligned}$$

Now the counts in front combine: five plus four minus two. $7\sqrt{2}$

Solution (b) — instructor-judged. The three radicands looked different but each is a perfect square times the same small number, so after simplifying all three terms carry the identical radical part and become like terms. Only then may the counts be added and

subtracted. An answer that jumps straight to combining, or that adds the radicands, does not earn credit.